Python Programming

Python · strings, loops & lists · Notes 1/5

READ THE QUESTION / STUDY THE ANSWER

Q01. [Coding] Write a program to reverse an input string.

```
s = input()
print(s[::-1])
```

Q02. [Coding] Write a program to check whether an input word is a palindrome.

```
s=input()
print('Palindrome' if s==s[::-1] else 'Not palindrome')
```

Q03. [Coding] Write a program to count vowels in an input string.

```
s=input().lower()
print(sum(1 for ch in s if ch in 'aeiou'))
```

Q04. [Coding] Write a program to print the multiplication table of an input integer from 1 to 10.

```
n=int(input())
for i in range(1,11):
    print(f'{n} x {i} = {n*i}')
```

Q05. [Coding] Write a program to read two integers and print their sum.

```
a = int(input())
b = int(input())
print(a + b)
```

Q06. [Coding] Write a program to check whether a number is prime.

```
n=int(input())
is_prime=n>1
for i in range(2,int(n**0.5)+1):
    if n%i==0:
        is_prime=False
        break
print('Prime' if is_prime else 'Not prime')
```

Python Programming

Python · strings, loops & lists · Notes 2/5

READ THE QUESTION / STUDY THE ANSWER

Q07. [Coding] Write a program to convert Celsius to Fahrenheit using F = C * 9/5 + 32.

```
c = float(input())
print(c * 9 / 5 + 32)
```

Q08. [Coding] Write a program to swap two variables using a temporary variable.

```
a=int(input())
b=int(input())
temp=a
a=b
b=temp
print(a,b)
```

Q09. [Coding] Write a program that prints whether an integer is even or odd.

```
n = int(input())
print('Even' if n % 2 == 0 else 'Odd')
```

Q10. [Coding] Write a program to count the number of words in an input sentence.

```
s=input().strip()
print(len(s.split()))
```

Q11. [Coding] Write a program to print the larger of two numbers.

```
a = int(input())
b = int(input())
print(max(a, b))
```

Q12. [Coding] Write a program to print the first n natural numbers and their sum.

```
n=int(input())
total=0
for i in range(1,n+1):
    print(i)
    total+=i
print('Sum:',total)
```

Python Programming

Python · strings, loops & lists · Notes 3/5

READ THE QUESTION / STUDY THE ANSWER

Q13. [Coding] Use a for loop to print numbers from 1 through 10.

```
for i in range(1, 11):
    print(i)
```

Q14. Which keyword defines a function? A) function B) define C) def D) fun

```
C. Functions are defined with def.
```

Q15. [Coding] Write a function square(n) that returns n multiplied by itself.

```
def square(n): return n * n
```

Q16. What is the output of 'Python'[0]? A) P B) y C) Python D) 0

```
A. String indexing begins at zero.
```

Q17. [Coding] Write a program to count the characters in an input string.

```
s = input()
print(len(s))
```

Q18. Which method adds an item to the end of a list? A) add() B) append() C) push() D) put()

```
B. append() adds one item at the end.
```

Python Programming

Python · strings, loops & lists · Notes 4/5

READ THE QUESTION / STUDY THE ANSWER

Q19. [Coding] Given a list of numbers, print its largest element.

```
nums = list(map(int, input().split()))
print(max(nums))
```

Q20. What is len([4, 5, 6])? A) 2 B) 3 C) 4 D) 6

```
B. The list has three elements.
```

Q21. [Coding] Write a program to calculate the sum of all elements in a list.

```
nums = list(map(int, input().split()))
print(sum(nums))
```

Q22. Which collection is immutable? A) list B) set C) tuple D) dictionary

```
C. Tuples cannot be changed by item assignment.
```

Q23. [Coding] Create a dictionary with keys 'name' and 'age', then print the name.

```
student = {'name': 'Ravi', 'age': 20}
print(student['name'])
```

Q24. What is the result of len({1, 2, 2, 3})? A) 2 B) 3 C) 4 D) 5

```
B. Sets store unique elements only.
```

Python Programming

Python · strings, loops & lists · Notes 5/5

READ THE QUESTION / STUDY THE ANSWER

Q25. [Coding] Write a program to count how many times a target value appears in a list.

```
nums = list(map(int, input().split()))
target = int(input())
print(nums.count(target))
```

Q26. What does the and operator do? A) True if either operand is true B) True only if both operands are true C) Reverses a condition D) Assigns a value

```
B. and requires both conditions to be true.
```

Q27. [Coding] Use try/except to handle division by zero when dividing two input numbers.

```
try:
    a=float(input())
    b=float(input())
    print(a/b)
except ZeroDivisionError:
    print('Cannot divide by zero')
```

Q28. Which is a valid variable name? A) 2name B) student-name C) student_name D) class

```
C. Names cannot start with a digit, contain hyphens, or be keywords.
```

Q29. [Coding] Write a program to reverse an input string.

```
s = input()
print(s[::-1])
```

Q30. [Coding] Write a program to print the factorial of a non-negative integer using a loop.

```
n=int(input())
result=1
for i in range(1,n+1):
    result*=i
print(result)
```

Java Programming

Java · methods, arrays & objects · Notes 1/5

READ THE QUESTION / STUDY THE ANSWER

Q01. [Coding] Write Java code to reverse a string str.

```
String reversed = new StringBuilder(str).reverse().toString();
```

Q02. [Coding] Write Java code to check whether a string is a palindrome.

```
String rev=new StringBuilder(s).reverse().toString();
System.out.println(s.equals(rev) ?
"Palindrome" : "Not palindrome");
```

Q03. [Coding] Write Java code to read two integers and print their sum.

```
Scanner sc = new Scanner(System.in);
int a=sc.nextInt(), b=sc.nextInt();
System.out.println(a+b);
```

Q04. [Coding] Write Java code to count vowels in a string s, ignoring case.

```
int count=0;
for(char c:s.toLowerCase().toCharArray())
    if("aeiou".indexOf(c)>=0) count++;
System.out.println(count);
```

Q05. [Coding] Write an if/else statement to print whether n is positive, negative, or zero.

```
if(n>0)
    System.out.println("Positive");
else if(n<0) System.out.println("Negative");
else
System.out.println("Zero");
```

Q06. [Coding] Write Java code to print the multiplication table of n from 1 to 10.

```
for(int i=1;i<=10;i++)
    System.out.println(n + " x " + i + " = " + (n*i));
```

Java Programming

Java · methods, arrays & objects · Notes 2/5

READ THE QUESTION / STUDY THE ANSWER

Q07. [Coding] Write a for loop to print numbers 1 to 10.

```
for(int i=1; i<=10; i++)
    System.out.println(i);
```

Q08. [Coding] Write Java code to check whether an integer n is prime.

```
boolean prime=n>1;
for(int i=2;i<=Math.sqrt(n);i++)
    if(n%i==0) {
    prime=false;
    break;
}
System.out.println(prime ? "Prime" : "Not prime");
```

Q09. [Coding] Write a method int square(int n) that returns n*n.

```
static int square(int n) {
    return n*n;
}
```

Q10. [Coding] Write Java code to swap two integer variables using a temporary variable.

```
int temp=a;
a=b;
b=temp;
```

Q11. [Coding] Write code to find the largest value in an int array named nums.

```
int max=nums[0];
for(int n: nums)
    if(n>max) max=n;
System.out.println(max);
```

Q12. [Coding] Write Java code to count words in a sentence s, treating whitespace as separators.

```
String trimmed=s.trim();
int count=trimmed.isEmpty()?0:trimmed.split("\\s+").length;
System.out.println(count);
```

Java Programming

Java · methods, arrays & objects · Notes 3/5

READ THE QUESTION / STUDY THE ANSWER

Q13. [Coding] Write code to calculate the sum of an int array.

```
int sum=0;
for(int n: nums)
    sum+=n;
System.out.println(sum);
```

Q14. [Coding] Write Java code to calculate the sum of integers from 1 to n using a loop.

```
int sum=0;
for(int i=1;i<=n;i++)
    sum+=i;
System.out.println(sum);
```

Q15. [Coding] Create a simple Student class with a name field and a constructor that sets it.

```
class Student {
    String name;
    Student(String name) {
        this.name=name;
    }
}
```

Q16. A constructor has: A) Any name B) Same name as class C) Name main D) Return type void

```
B. A constructor shares the class name and has no return type.
```

Q17. [Coding] Write code to reverse a string str using a loop or StringBuilder.

```
String reversed = new StringBuilder(str).reverse().toString();
```

Q18. Which block handles an exception? A) try B) catch C) throw D) final

```
B. catch handles a matching exception from try.
```

Java Programming

Java · methods, arrays & objects · Notes 4/5

READ THE QUESTION / STUDY THE ANSWER

Q19. [Coding] Write a program fragment to print whether an integer is even or odd.

```
System.out.println(n%2==0 ? "Even" : "Odd");
```

Q20. Which class is commonly used for keyboard input? A) Printer B) Scanner C) Input D) ReaderOnly

```
B. java.util.Scanner reads input from System.in.
```

Q21. [Coding] Write a while loop that prints 5, 4, 3, 2, 1.

```
int i=5;
while(i>=1) {
    System.out.println(i);
    i--;
}
```

Q22. What is the default value of an instance int field? A) 1 B) -1 C) 0 D) null

```
C. Instance int fields default to zero.
```

Q23. [Coding] Write code to count the number of elements in an array nums.

```
System.out.println(nums.length);
```

Q24. Which keyword refers to the current object? A) super B) this C) self D) current

```
B. this refers to the current instance.
```

Java Programming

Java · methods, arrays & objects · Notes 5/5

READ THE QUESTION / STUDY THE ANSWER

Q25. [Coding] Write code to convert the string "123" to an integer.

```
int value = Integer.parseInt("123");
```

Q26. Which operator means logical AND? A) & only B) && C) || D) !

```
B. && is the short-circuit logical AND operator.
```

Q27. [Coding] Write a method that returns the average of three double values.

```
static double average(double a,double b,double c) {
    return (a+b+c)/3.0;
}
```

Q28. Which access modifier allows access from any class, subject to module/package rules? A) private B) public C) protected only D) local

```
B. public is the broadest standard access modifier.
```

Q29. [Coding] Write a switch statement that prints a weekday name for an integer 1 to 3.

```
switch(day) {
    case 1:System.out.println("Monday");
    break;
    case
    2:System.out.println("Tuesday");
    break;
    case 3:System.out.println("Wednesday");
    break;
    default:System.out.println("Invalid");
}
```

Q30. [Coding] Write code to find the number of vowels in a string s, ignoring case.

```
int count=0;
for(char c:s.toLowerCase().toCharArray())
    if("aeiou".indexOf(c)>=0) count++;
System.out.println(count);
```

C Programming

C · pointers, strings & functions · Notes 1/5

READ THE QUESTION / STUDY THE ANSWER

Q01. [Coding] Write a C program fragment to reverse a string stored in char s[].

```
int len=strlen(s);
for(int i=len-1;i>=0;i--)
    putchar(s[i]);
```

Q02. [Coding] Write C code to check whether a string s is a palindrome.

```
int i=0,j=strlen(s)-1,ok=1;
while(i<j) {
    if(s[i]!=s[j]) {
        ok=0;
        break;
    }
    i++;
    j--;
}
puts(ok?"Palindrome":"Not palindrome");
```

Q03. [Coding] Write a C program to read two integers and print their sum.

```
#include <stdio.h>
int main(void) {
    int a,b;
    scanf("%d%d",&a,&b);
    printf("%d",a+b);
    return 0;
}
```

Q04. [Coding] Write C code to count vowels in a null-terminated string s.

```
int count=0;
for(int i=0;s[i]!='\0';i++) {
    char c=tolower((unsigned char)s[i]);
    if(c=='a'||c=='e'||c=='i'||c=='o'||c=='u')
        count++;
}
printf("%d",count);
```

Q05. [Coding] Write a program to determine whether an integer is even or odd.

```
if(n%2==0)
    printf("Even");
else printf("Odd");
```

Q06. [Coding] Write C code to print the multiplication table of n from 1 to 10.

```
for(int i=1;i<=10;i++)
    printf("%d x %d = %d\n",n,i,n*i);
```

C Programming

C · pointers, strings & functions · Notes 2/5

READ THE QUESTION / STUDY THE ANSWER

Q07. [Coding] Write a for loop that prints integers from 1 through 10.

```
for(int i=1;i<=10;i++)
    printf("%d ",i);
```

Q08. [Coding] Write C code to check whether n is prime.

```
int prime=n>1;
for(int i=2;i*i<=n;i++)
    if(n%i==0) {
    prime=0;
    break;
}
puts(prime?"Prime":"Not
prime");
```

Q09. [Coding] Write a C function that returns the square of an integer.

```
int square(int n) {
    return n*n;
}
```

Q10. [Coding] Write C code to swap two integers a and b using a temporary variable.

```
int temp=a;
a=b;
b=temp;
```

Q11. [Coding] Write code to find the largest element in an integer array a of length n.

```
int max=a[0];
for(int i=1;i<n;i++)
    if(a[i]>max) max=a[i];
printf("%d",max);
```

Q12. [Coding] Write C code to count words separated by spaces in a string s (assume words are separated by single spaces).

```
int count=0;
for(int i=0;s[i]!='\0';i++)
    if(s[i]!=' ' && (i==0 || s[i-1]==' ')) count++;
printf("%d",count);
```

C Programming

C · pointers, strings & functions · Notes 3/5

READ THE QUESTION / STUDY THE ANSWER

Q13. [Coding] Declare an integer variable x and a pointer p that stores its address.

```
int x=10;
int *p=&x;
```

Q14. [Coding] Write C code to calculate the sum of integers from 1 to n using a loop.

```
int sum=0;
for(int i=1;i<=n;i++)
    sum+=i;
printf("%d",sum);
```

Q15. [Coding] Write code to calculate the sum of n integers stored in array a.

```
int sum=0;
for(int i=0;i<n;i++)
    sum+=a[i];
printf("%d",sum);
```

Q16. Which keyword declares a structure? A) object B) struct C) record D) class

```
B. C uses struct to define a structure type.
```

Q17. [Coding] Write a C string declaration that can hold the word "Hello" including its terminator.

```
char text[6] = "Hello";
```

Q18. Which function reads formatted input? A) printf() B) scanf() C) puts() D) strlen()

```
B. scanf() reads formatted input.
```

C Programming

C · pointers, strings & functions · Notes 4/5

READ THE QUESTION / STUDY THE ANSWER

Q19. [Coding] Write a switch statement for choices 1 and 2, printing Yes and No respectively.

```
switch(choice) {
    case 1: printf("Yes");
    break;
    case 2: printf("No");
    break;
    default:
    printf("Invalid");
}
```

Q20. What does sizeof(x) return? A) The value of x B) Size in bytes C) Address of x D) Number of digits

```
B. sizeof gives the size in bytes of a type or object.
```

Q21. [Coding] Write code to swap two integers using a temporary variable.

```
int temp=a;
a=b;
b=temp;
```

Q22. Which library function returns a string's length? A) size() B) strlen() C) length() D) strsize()

```
B. strlen() is declared in string.h.
```

Q23. [Coding] Write a loop that counts down from 5 to 1.

```
for(int i=5;i>=1;i--)
    printf("%d ",i);
```

Q24. What is a pointer? A) A loop B) A variable that stores an address C) A keyword D) A data type only for text

```
B. A pointer stores a memory address.
```

C Programming

C · pointers, strings & functions · Notes 5/5

READ THE QUESTION / STUDY THE ANSWER

Q25. [Coding] Write code to compute the average of three integers as a floating-point result.

```
float avg=(a+b+c)/3.0f;
printf("%.2f",avg);
```

Q26. Which is a valid C identifier? A) 2total B) total marks C) total_marks D) int

```
C. Identifiers may contain underscores but cannot be keywords or start with digits.
```

Q27. [Coding] Write a function prototype for a function named add that accepts two ints and returns an int.

```
int add(int a, int b);
```

Q28. Which operator accesses a structure member through a structure variable? A) -> B) . C) :: D) &

```
B. The dot operator accesses members through a structure object.
```

Q29. [Coding] Write code to print each character of a null-terminated string s on a separate line.

```
for(int i=0;s[i]!='\0';i++)
    printf("%c\n",s[i]);
```

Q30. [Coding] Write a program fragment to test whether n is positive, negative, or zero.

```
if(n>0)
    puts("Positive");
else if(n<0) puts("Negative");
else puts("Zero");
```

SQL

SQL · query patterns · Notes 1/3

READ THE QUESTION / STUDY THE ANSWER

Q01. In a database table, a row usually represents: A) An attribute B) A record C) A database D) A query

```
B. A row represents one record.
```

Q02. Which key uniquely identifies each row in a table? A) Foreign key B) Primary key C) Duplicate key D) Sort key

```
B. A primary key uniquely identifies rows.
```

Q03. [Coding] Write SQL to create a Students table with id integer primary key and name VARCHAR(50).

```
CREATE TABLE Students (
    id INT PRIMARY KEY,
    name VARCHAR(50)
);
```

Q04. [Coding] Insert student id 1 and name 'Ravi' into Students(id, name).

```
INSERT INTO Students (id, name)
    VALUES (1, 'Ravi');
```

Q05. [Coding] Select all columns from a table named Students.

```
SELECT *
    FROM Students;
```

Q06. [Coding] Select only names of students whose age is greater than 18.

```
SELECT name
    FROM Students
    WHERE age > 18;
```

SQL

SQL · query patterns · Notes 2/3

READ THE QUESTION / STUDY THE ANSWER

Q07. Which command changes existing row values? A) UPDATE B) ALTER C) CREATE D) SELECT

```
A. UPDATE modifies rows in a table.
```

Q08. [Coding] Increase salary by 1000 for employees in department 'IT'.

```
UPDATE Employees
    SET salary = salary + 1000
    WHERE department = 'IT';
```

Q09. Which command removes selected rows from a table? A) DROP B) DELETE C) REMOVE TABLE D) CLEAR

```
B. DELETE removes rows; use WHERE to restrict which rows.
```

Q10. [Coding] Delete the student whose id is 5.

```
DELETE
    FROM Students
    WHERE id = 5;
```

Q11. Which clause sorts query results? A) GROUP BY B) ORDER BY C) SORT D) ARRANGE

```
B. ORDER BY sorts result rows.
```

Q12. [Coding] Show unique department values from Employees.

```
SELECT DISTINCT department
    FROM Employees;
```

SQL

READ THE QUESTION / STUDY THE ANSWER

Q13. Which aggregate function counts rows? A) SUM() B) AVG() C) COUNT() D) TOTAL()

```
C. COUNT() counts rows or non-NULL values, depending on its argument.
```

Q14. [Coding] Find the average salary from Employees.

```
SELECT AVG(salary)
    FROM Employees;
```

Q15. [Coding] Count employees in each department.

```
SELECT department, COUNT(*)
    FROM Employees
    GROUP BY department;
```

Q16. A foreign key is mainly used to: A) Format text B) Link related tables C) Sort rows automatically D) Encrypt a table

```
B. Foreign keys establish relationships and enforce referential integrity.
```

Q17. [Coding] Join Students(sid, name) with Marks(sid, score) to show each student's name and score.

```
SELECT Students.name, Marks.score
    FROM Students
    JOIN Marks
    ON Students.sid = Marks.sid;
```

JavaScript

READ THE QUESTION / STUDY THE ANSWER

Q01. [Coding] Write JavaScript to reverse a string s.

```
console.log(s.split('').reverse().join(''));
```

Q02. [Coding] Write JavaScript to check whether a string s is a palindrome.

```
console.log(s===s.split('').reverse().join('') ? 'Palindrome' : 'Not palindrome');
```

Q03. [Coding] Write JavaScript to print Hello, World! to the console.

```
console.log('Hello, World!');
```

Q04. [Coding] Write JavaScript to count vowels in a string s.

```
const count=(s.match(/[aeiou]/gi)||[]).length;
console.log(count);
```

Q05. [Coding] Write code to add two numbers a and b and print the result.

```
console.log(a + b);
```

Q06. [Coding] Write JavaScript to print the multiplication table of n from 1 to 10.

```
for(let i=1;i<=10;i++)
    console.log(`${n} x ${i} = ${n*i}`);
```

JavaScript

READ THE QUESTION / STUDY THE ANSWER

Q07. [Coding] Write an if/else statement that prints Even or Odd for n.

```
if (n % 2 === 0)
    console.log('Even');
else console.log('Odd');
```

Q08. [Coding] Write JavaScript to check whether a positive integer n is prime.

```
let prime=n>1;
for(let i=2;i<=Math.sqrt(n);i++)
    if(n%i===0) {
    prime=false;
    break;
}
console.log(prime?'Prime':'Not prime');
```

Q09. [Coding] Write a for loop that prints numbers 1 through 5.

```
for (let i=1; i<=5; i++)
    console.log(i);
```

Q10. [Coding] Write JavaScript to swap values a and b using a temporary variable.

```
let temp=a;
a=b;
b=temp;
console.log(a,b);
```

Q11. [Coding] Write a function multiply(a, b) that returns their product.

```
function multiply(a, b) {
    return a * b;
}
```

Q12. [Coding] Write JavaScript to count words in a sentence s separated by whitespace.

```
const trimmed=s.trim();
console.log(trimmed ? trimmed.split(/\s+/).length : 0);
```

JavaScript

READ THE QUESTION / STUDY THE ANSWER

Q13. [Coding] Create an array of three fruits and print its first element.

```
const fruits=['apple','banana','mango'];
console.log(fruits[0]);
```

Q14. [Coding] Write JavaScript to calculate the sum of integers from 1 to n using a loop.

```
let sum=0;
for(let i=1;i<=n;i++)
    sum+=i;
console.log(sum);
```

Q15. [Coding] Write code to find the sum of numbers in array nums.

```
let sum=0;
for (const n of nums)
    sum+=n;
console.log(sum);
```

Q16. How do you access the name property of object person? A) person->name B) person.name C) person::name D) person(name)

```
B. Dot notation accesses object properties.
```

Q17. [Coding] Create an object person with name 'Asha' and age 19.

```
const person = {
    name: 'Asha', age: 19
}
;
```

Q18. What does 'hello'.toUpperCase() return? A) hello B) HELLO C) Hello D) Error

```
B. toUpperCase() returns an uppercase string.
```

JavaScript

READ THE QUESTION / STUDY THE ANSWER

Q19. [Coding] Write code to print the length of string text.

```
console.log(text.length);
```

Q20. Which loop is commonly used to iterate while a condition remains true? A) if B) while C) switch D) case

```
B. while repeats while its condition is true.
```

Q21. [Coding] Write a while loop that prints 1, 2, and 3.

```
let i=1;
while(i<=3) {
    console.log(i);
    i++;
}
```

Q22. What is the result of Boolean(0)? A) true B) false C) 0 D) undefined

```
B. Zero is falsy in JavaScript.
```

Q23. [Coding] Write an arrow function named square that returns n*n.

```
const square = n => n * n;
```

Q24. Which browser API selects an element by its ID? A) document.getElementById() B) console.get() C) window.findId() D) html.selectId()

```
A. getElementById() selects the element with the specified ID.
```

JavaScript

JavaScript · language & browser basics · Notes 5/5

READ THE QUESTION / STUDY THE ANSWER

Q25. [Coding] Write JavaScript to change the text of element id 'message' to 'Welcome'.

```
document.getElementById('message').textContent = 'Welcome';
```

Q26. Which event commonly occurs when a user clicks a button? A) hoveronly B) click C) submittext D) presskey

```
B. The click event responds to mouse or equivalent activation.
```

Q27. [Coding] Show how to attach a click event listener to button element btn.

```
btn.addEventListener('click', () => console.log('Clicked'));
```

Q28. What does console.log() do? A) Creates an HTML element B) Prints information to the console C) Opens a database D) Stops a loop

```
B. console.log() writes a message to the developer console.
```

Q29. [Coding] Write code to calculate the largest of two numbers a and b.

```
console.log(a > b ? a : b);
```

DSA using Java/C

DSA · structures & algorithms · Notes 1/4

READ THE QUESTION / STUDY THE ANSWER

Q01. [Coding] Write a simple linear search in Java/C that returns the index of target or -1 if absent.

```
for(int i=0;i<n;i++)
    if(a[i]==target) return i;
return -1;
```

Q02. [Coding] Write pseudocode to find the minimum element in an integer array.

```
min=a[0];
for each remaining element x: if x<min, set min=x.
```

Q03. [Coding] Write pseudocode or Java/C code for linear search for target in an array.

```
for (i=0; i<n; i++)
    if (a[i]==target) return i;
return -1;
```

Q04. [Coding] Write the steps to push an item onto a stack implemented with an array.

```
If space remains, increment top and store the item at stack[top].
```

Q05. [Coding] In a sorted ascending array, describe the basic binary-search step after comparing target with a[mid].

```
If target < a[mid], set high=mid-1;
if target > a[mid], set low=mid+1;
otherwise found.
```

Q06. [Coding] Write the steps to enqueue an item in a simple array-based queue.

```
If space remains, store the item at the rear position and advance rear.
```

DSA using Java/C

READ THE QUESTION / STUDY THE ANSWER

Q07. [Coding] Show one bubble-sort pass on [3, 1, 2] in ascending order.

```
Compare 3 and 1 -> [1,3,2];
compare 3 and 2 -> [1,2,3].
```

Q08. [Coding] Write the basic code/steps to swap adjacent elements if they are out of ascending order.

```
if(a[j]>a[j+1]) {
    temp=a[j];
    a[j]=a[j+1];
    a[j+1]=temp;
}
```

Q09. [Coding] Name the two common stack operations and what they do.

```
push adds an item to the top;
pop removes the top item.
```

Q10. A queue follows which rule? A) LIFO B) FIFO C) Largest-first D) Smallest-first

```
B. First in, first out.
```

Q11. [Coding] If 10 then 20 are enqueued, what is removed first by dequeue?

```
10, because the queue is FIFO.
```

Q12. In a singly linked list, each node usually stores: A) Data and a next reference/pointer B) Only an index C) A SQL query D) Only a file name

```
A. Each node holds data and a link to the next node.
```

DSA using Java/C

READ THE QUESTION / STUDY THE ANSWER

Q13. [Coding] Write the basic steps to insert a node at the beginning of a singly linked list.

```
new.next=head;
head=new.
```

Q14. Which operation adds an item to a stack? A) pop B) push C) dequeue D) search

```
B. push adds an item to the stack.
```

Q15. Which operation removes an item from a queue? A) enqueue B) push C) dequeue D) append-only

```
C. dequeue removes the front item.
```

Q16. What is the usual time complexity of accessing an array element by index? A) O(1) B) O(n) C) O(n²) D) O(log n) always

```
A. Indexed array access is constant time.
```

Q17. What is the worst-case time complexity of linear search in n elements? A) O(1) B) O(log n) C) O(n) D) O(n²)

```
C. It may inspect every element.
```

Q18. [Coding] Trace linear search for target 7 in [2, 4, 7, 9]. Which index is returned with zero-based indexing?

```
Index 2.
```

DSA using Java/C

READ THE QUESTION / STUDY THE ANSWER

Q19. Which structure is useful for undo operations? A) Stack B) Queue C) Sorted array only D) Graph only

```
A. Undo commonly reverses the most recent action first.
```

Q20. Which is the simplest description of O(n)? A) Constant work regardless of input size B) Work grows roughly in proportion to input size C) Work grows as n squared D) Work halves every step

```
B. Linear time grows proportionally to the number of elements.
```

Operating Systems

Operating systems · core concepts · Notes 1/3

READ THE QUESTION / STUDY THE ANSWER

Q01. What is the main role of an operating system? A) Manage hardware and software resources B) Only edit photos C) Only browse websites D) Replace the CPU

```
A. The OS manages resources and provides services to programs.
```

Q02. Which is an example of an operating system? A) Python B) Linux C) HTML D) SQL

```
B. Linux is an operating system.
```

Q03. A process is: A) A program in execution B) A keyboard key C) A file extension D) A network cable

```
A. A process is a running instance of a program.
```

Q04. A thread is generally: A) A unit of execution within a process B) A storage disk C) A type of monitor D) A network protocol

```
A. Threads are execution paths within a process.
```

Q05. Which scheduling algorithm serves processes in arrival order? A) FCFS B) LRU C) Paging D) DNS

```
A. First-Come, First-Served schedules in arrival order.
```

Q06. Which scheduling method uses a fixed time slice? A) Round Robin B) FCFS only C) FIFO page replacement D) Segmentation

```
A. Round Robin assigns each process a time quantum.
```

Operating Systems

Operating systems · core concepts · Notes 2/3

READ THE QUESTION / STUDY THE ANSWER

Q07. RAM is usually considered: A) Volatile memory B) Permanent optical storage C) A CPU register only D) A network address

```
A. RAM typically loses its contents when power is removed.
```

Q08. Paging divides memory into fixed-size: A) Pages and frames B) Files and folders C) Packets and ports D) Processes and users

```
A. Virtual memory pages map to physical memory frames.
```

Q09. Virtual memory allows a system to: A) Use secondary storage to extend the apparent memory space B) Eliminate all storage C) Increase CPU clock speed directly D) Replace the OS

```
A. Virtual memory lets programs use an address space larger than physical RAM.
```

Q10. Which is a common file operation? A) Open B) Compile only C) Route D) Ping

```
A. Opening a file is a standard file-system operation.
```

Q11. A deadlock can occur when processes: A) Wait indefinitely for resources held by each other B) Finish immediately C) Use no resources D) Only print output

```
A. Circular waiting for resources can prevent progress.
```

Q12. Which is a type of operating system? A) Real-time OS B) Spreadsheet OS only C) SQL OS D) HTML OS

```
A. Real-time operating systems are designed to meet timing constraints.
```

Operating Systems

Operating systems · core concepts · Notes 3/3

READ THE QUESTION / STUDY THE ANSWER

Q13. What does context switching do? A) Switches CPU execution from one process/thread to another B) Deletes every file C) Formats a hard drive D) Converts RAM into ROM

```
A. The OS saves and restores execution state during a switch.
```

Q14. Which is an example of secondary storage? A) SSD B) CPU cache only C) Register D) ALU

```
A. An SSD stores data persistently.
```

Q15. A mutex is commonly used to: A) Protect shared resources from simultaneous access B) Resolve DNS names C) Compress files D) Assign IP addresses

```
A. A mutex provides mutual exclusion around critical sections.
```

Computer Networks

Computer networks · protocols & devices · Notes 1/3

READ THE QUESTION / STUDY THE ANSWER

Q01. A LAN usually covers: A) A small local area such as a building B) The entire planet only C) Only satellites D) A single CPU register

```
A. A LAN connects devices within a limited area.
```

Q02. Which topology connects devices to a central switch/hub? A) Star B) Ring only C) Bus only D) Mesh only

```
A. In a star topology, devices connect to a central node.
```

Q03. How many layers are in the OSI model? A) 4 B) 5 C) 7 D) 8

```
C. The OSI reference model has seven layers.
```

Q04. Which protocol is used to load secure web pages? A) HTTPS B) FTP only C) ARP only D) SMTP only

```
A. HTTPS is HTTP protected by TLS.
```

Q05. DNS mainly translates: A) Domain names to IP addresses B) MAC addresses to passwords C) Files to folders D) Images to text

```
A. DNS resolves domain names to records such as IP addresses.
```

Q06. An IP address is used to: A) Identify a network interface logically for communication B) Store a file's contents C) Name a keyboard D) Replace a router

```
A. IP addressing supports routing packets across networks.
```

Computer Networks

Computer networks · protocols & devices · Notes 2/3

READ THE QUESTION / STUDY THE ANSWER

Q07. A MAC address is associated with: A) A network interface at the link layer B) A website title C) A database row D) A CPU instruction

```
A. MAC addresses identify interfaces on a local link.
```

Q08. Which device forwards packets between networks? A) Router B) Hub C) Monitor D) Keyboard

```
A. Routers forward IP packets between networks.
```

Q09. A switch commonly forwards Ethernet frames using: A) MAC addresses B) Domain names only C) File names D) CPU temperatures

```
A. A switch learns MAC addresses to forward frames.
```

Q10. A hub generally sends incoming data to: A) All its other ports B) Only the destination port using a MAC table C) A DNS server only D) No ports

```
A. A hub repeats signals to all connected ports.
```

Q11. Which transport protocol is connection-oriented and reliable? A) TCP B) UDP C) ARP D) DNS

```
A. TCP provides reliable, ordered delivery.
```

Q12. Which protocol is commonly used to send email between mail servers? A) SMTP B) HTTP C) DHCP D) ICMP only

```
A. SMTP is used for sending email.
```

Computer Networks

Computer networks · protocols & devices · Notes 3/3

READ THE QUESTION / STUDY THE ANSWER

Q13. What does DHCP commonly provide to a host? A) Network configuration such as an IP address B) Website source code C) A MAC address burned into hardware D) A CPU process

```
A. DHCP can assign IP addresses and related configuration.
```

Q14. The TCP/IP model is commonly described using how many layers in its four-layer version? A) 2 B) 4 C) 7 D) 9

```
B. The common four-layer TCP/IP model has link, internet, transport, and application layers.
```

Q15. What does bandwidth broadly describe in networking? A) Data-carrying capacity per unit time B) Physical cable color C) Number of keyboards D) A domain name

```
A. Bandwidth describes the capacity of a communication channel.
```